

$$e_{\ell}^{1, \bar{1}, 2, i_3 - i_4} \rightarrow$$
$$\begin{aligned}& \frac{1}{16\pi^2} \left(\frac{2}{27} \sum_p g_1^4 LF_{3,0}[m_d^{-P}] \delta_{i1i2} \delta_{i3i4} - \frac{5}{36} \sum_p g_1^4 LF_{4,-1}[m_d^{-P}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{135} \sum_p g_1^4 LF_{5,-2}[m_d^{-P}] \right. \\& \quad \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} \sum_p g_1^4 LF_{3,0}[m_e^{-P}] \delta_{i1i2} \delta_{i3i4} - \frac{5}{12} \sum_p g_1^4 LF_{4,-1}[m_e^{-P}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{8}{45} \sum_p g_1^4 LF_{5,-2}[m_e^{-P}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{9} \sum_p g_1^4 LF_{3,0}[m_l^{-P}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{5}{24} \sum_p g_1^4 LF_{4,-1}[m_l^{-P}] \delta_{i1i2} \delta_{i3i4} + \frac{4}{45} \sum_p g_1^4 LF_{5,-2}[m_l^{-P}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{27} \sum_p g_1^4 LF_{3,0}[m_q^{-P}] \delta_{i1i2} \delta_{i3i4} - \frac{5}{72} \sum_p g_1^4 LF_{4,-1}[m_q^{-P}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{4}{135} \sum_p g_1^4 LF_{5,-2}[m_q^{-P}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{27} \sum_p g_1^4 LF_{3,0}[m_u^{-P}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{5}{9} \sum_p g_1^4 LF_{4,-1}[m_u^{-P}] \delta_{i1i2} \delta_{i3i4} + \frac{32}{135} \sum_p g_1^4 LF_{5,-2}[m_u^{-P}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{9} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{6} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{8}{45} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{12} g_1^4 LF_{2,1,0}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^4 LF_{2,2,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{1}{6} g_1^4 LF_{3,1,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^4 LF_{4,1,-2}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{12} g_1^4 LF_{2,1,0}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^4 LF_{2,2,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{1}{6} g_1^4 LF_{3,1,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^4 LF_{4,1,-2}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{48} g_1^4 LF_{2,1,0}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{48} g_1^4 LF_{2,2,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{1}{24} g_1^4 LF_{3,1,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{48} g_1^4 LF_{4,1,-2}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{48} g_1^4 LF_{2,1,0}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{48} g_1^4 LF_{2,2,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{1}{24} g_1^4 LF_{3,1,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{48} g_1^4 LF_{4,1,-2}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{16} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{16} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{1}{8} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{16} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{16} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{16} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{1}{8} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{16} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\& \quad \frac{1}{3} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} LF_{2,1,0}[m_e^{-P}, \tilde{\mu}] \delta_{i3i4} + \frac{1}{6} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} LF_{2,2,-1}[m_e^{-P}, \tilde{\mu}] \delta_{i3i4} + \\& \quad \frac{1}{6} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} LF_{3,1,-1}[m_e^{-P}, \tilde{\mu}] \delta_{i3i4} - \frac{1}{6} g_1^4 LF_{2,1,0}[m_e^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{12} g_1^4 LF_{3,1,-1}[m_e^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{6} g_1^4 LF_{2,1,0}[m_e^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{12} g_1^4 LF_{3,1,-1}[m_e^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{2} \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_e^{pr} a_e^{pr} LF_{3,1,0}[m_l^{-P}, m_e^{-r}] + \\& \quad \frac{3}{2} \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_e^{pr} a_e^{pr} LF_{4,1,-1}[m_l^{-P}, m_e^{-r}] - \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_e^{pr} a_e^{pr} LF_{5,1,-2}[m_l^{-P}, m_e^{-r}] - \\& \quad \frac{1}{6} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{2,1,0}[m_l^{-P}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{12} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{2,2,-1}[m_l^{-P}, \tilde{\mu}] \delta_{i1i2} + \\& \quad \frac{1}{12} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{3,1,-1}[m_l^{-P}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{24} g_1^4 LF_{2,1,0}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{48} g_1^4 LF_{3,1,-1}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{8} g_1^2 g_2^2 LF_{2,1,0}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{16} g_1^2 g_2^2 LF_{3,1,-1}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^4 LF_{2,1,0}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{48} g_1^4 LF_{3,1,-1}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{8} g_1^2 g_2^2 LF_{2,1,0}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} + \\& \quad \frac{1}{16} g_1^2 g_2^2 LF_{3,1,-1}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{3}{2} \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_d^{pr} a_d^{pr} LF_{3,1,0}[m_q^{-P}, m_d^{-r}] + \\& \quad \frac{9}{2} \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_d^{pr} a_d^{pr} LF_{4,1,-1}[m_q^{-P}, m_d^{-r}] - 3 \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_d^{pr} a_d^{pr} LF_{5,1,-2}[m_q^{-P}, m_d^{-r}] - \\& \quad \frac{3}{2} \tilde{\mu}^2 \overline{y}_e^{i2i3} y_e^{i1i4} \overline{y}_u^{pr} y_u^{pr} LF_{3,1,0}[m_u^{-r}, m_q^{-P}] + \frac{9}{2} \tilde{\mu}^2 \overline{y}_e^{i2i3} y_e^{i1i4} \overline{y}_u^{pr} y_u^{pr} LF_{4,1,-1}[m_u^{-r}, m_q^{-P}] - \\& \quad 3 \tilde{\mu}^2 \overline{y}_e^{i2i3} y_e^{i1i4} \overline{y}_u^{pr} y_u^{pr} LF_{5,1,-2}[m_u^{-r}, m_q^{-P}] + g_1^2 \overline{y}_e^{i2i3} y_e^{i1i4} LF_{3,1,-1}[\tilde{\mu}, m_1] - \\& \quad 2 g_1^2 \overline{y}_e^{i2i3} y_e^{i1i4} LF_{4,1,-2}[\tilde{\mu}, m_1] + g_1^2 \overline{y}_e^{i2i3} y_e^{i1i4} LF_{5,1,-3}[\tilde{\mu}, m_1] + \\& \quad 3 g_2^2 \overline{y}_e^{i2i3} y_e^{i1i4} LF_{3,1,-1}[\tilde{\mu}, m_2] - 6 g_2^2 \overline{y}_e^{i2i3} y_e^{i1i4} LF_{4,1,-2}[\tilde{\mu}, m_2] + \\& \quad 3 g_2^2 \overline{y}_e^{i2i3} y_e^{i1i4} LF_{5,1,-3}[\tilde{\mu}, m_2] + \frac{1}{6} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} LF_{2,1,0}[\tilde{\mu}, m_e^{-P}] \delta_{i3i4}$$